MT-Wave

Wi-Fi and Non-Wi-Fi Interference Generator



MT-Wave is a compact RF interference generator for stress-testing wireless devices. It contains **two independent radios**: one generates real Wi-Fi traffic on the 2.4, 5 and 6 GHz bands, the other plays Bluetooth, Zigbee, DFS radar and AWGN signals.

Both radios can transmit at the same time, so one unit builds a combined interference scene that would otherwise need several signal sources.

Channel, bandwidth, data rate, power, traffic class and inter-frame spacing, down to **0 µs (back-to-back frames)**, are set from a built-in web control panel or over an HTTP API, so every test condition can be repeated exactly.

WI-FI BANDS	WI-FI OUTPUT LEVEL	INTER-FRAME SPACING	DFS RADAR LIBRARY
2.4 / 5 / 6 GHz	−30 to +15 dBm	16 to 3000 µs	8 regions

Key features

- **Two independent radios**: Wi-Fi and non-Wi-Fi interference from one unit, at the same time.
- **Real 802.11 traffic** on 2.4, 5 and 6 GHz: 802.11a/b/g/n/ac/ax, 20 / 40 / 80 MHz, one chain or 2×2 MIMO.
- **Precise level control**: Wi-Fi output from −30 to +15 dBm in 1 dB steps, ±0.5 dB accuracy.
- **Adjustable channel load**: inter-frame spacing from 16 µs (back-to-back) to 3000 µs, with WMM traffic classes.
- **Bluetooth and Zigbee**: Bluetooth Classic, Low Energy and Long Range; IEEE 802.15.4 frames.
- **DFS radar patterns** for eight regulatory regions, placed on any 5 GHz channel.
- **AWGN** from 2 to 20 MHz wide, tunable from 2.4 to 6 GHz.
- **Browser control panel and HTTP API**: nothing to install, and every setting can be scripted.

Applications

- **Wi-Fi robustness and QoS**: access point and client behaviour under co-channel and adjacent-channel congestion.
- **Wi-Fi / Bluetooth coexistence** in the 2.4 GHz band.
- **Zigbee, Thread and Matter coexistence** with busy Wi-Fi channels.
- **DFS development and pre-compliance**: radar detection, channel availability check and channel-move behaviour.
- **6 GHz and Wi-Fi 6E stress testing** across U-NII-5 to U-NII-8.
- **Sensitivity and SNR curves**: raise the noise floor with AWGN without moving the device under test.
- **Regression and field-issue reproduction**: scripted, repeatable interference scenarios.

Wi-Fi interference

The Wi-Fi radio transmits continuous 802.11 traffic that the unit generates itself. Placing it on the same channel as the device under test, or on a neighbouring one, produces co-channel or adjacent-channel interference at a level and channel load you choose.

Parameter	Specification
Standards	IEEE 802.11a / b / g / n / ac / ax (Wi-Fi 6 and Wi-Fi 6E)
Bands	2.4 GHz, 5 GHz and 6 GHz
Channel bandwidth	802.11a / b / g: 20 MHz 802.11n: 20 or 40 MHz 802.11ac / ax: 20, 40 or 80 MHz
Data rate (MCS)	802.11n: MCS 0 to 7 802.11ac: MCS 0 to 9 802.11ax: MCS 0 to 11
Transmit chains	Chain 1, Chain 2, or both chains (2×2 MIMO)
Output level	−30 to +15 dBm in 1 dB steps. Lower levels with an external attenuator.
Level accuracy	±0.5 dB
Traffic class (WMM)	Voice, Video, Best effort, Background, plus Maximum and Custom
Payload	3080 bytes with the preset traffic classes; 64 to 4000 bytes with Custom
Inter-frame spacing	0 to 3000 µs in 1 µs steps, added to the spacing of the selected traffic class. Custom at 0 µs gives back-to-back frames.
Transmission	Continuous until stopped, with a running timer
RF output	RF1 (front panel)

Wi-Fi channel coverage

Band	Standards	Channels	Centre frequencies
2.4 GHz	802.11b / g / n / ax	1 to 11	2412 to 2462 MHz
5 GHz	802.11a / n / ac / ax	36 to 64 (U-NII-1, U-NII-2A) 100 to 144 (U-NII-2C) 149 to 165 (U-NII-3)	5180 to 5320 MHz 5500 to 5720 MHz 5745 to 5825 MHz
6 GHz	802.11ax	1 to 233, every fourth channel (U-NII-5 to U-NII-8)	5955 to 7115 MHz

The 5 GHz list has 25 channels, including the DFS channels in U-NII-2A and U-NII-2C. Together with the radar generator, these let you check radar detection and channel-move behaviour on the channel the device is using.

Non-Wi-Fi interference

The second radio plays ready-made recordings of Bluetooth, Zigbee, radar and noise signals. You choose which signal to play and where to place it in the spectrum. One signal type is active at a time, and it runs independently of the Wi-Fi radio.

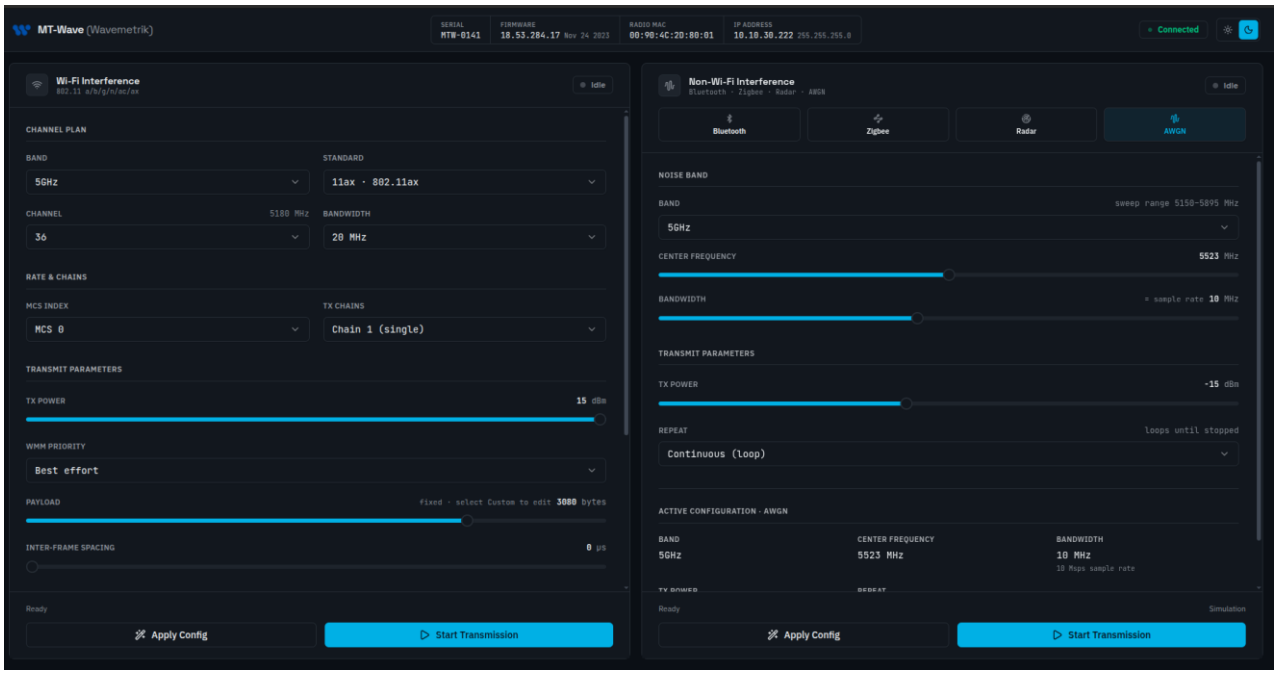
Parameter	Specification
Bluetooth	
Signal types	Bluetooth Classic: BR 1 Mbps (GFSK), EDR 2 Mbps (π/4-DQPSK), EDR 3 Mbps (8DPSK) Bluetooth Low Energy: LE 1M and LE 2M advertising BLE Long Range: LE Coded PHY, S=2 and S=8
Channel	0 to 78 (2402 to 2480 MHz in 1 MHz steps)
Zigbee (IEEE 802.15.4)	
Frame types	Data, Beacon, Ack, MAC Command
Channel	Zigbee channels 11 to 26 (2405 to 2480 MHz in 5 MHz steps) Also placeable on any 5 GHz Wi-Fi channel, 36 to 165
Radar (DFS test patterns)	
Regions	FCC, ETSI, Japan, Korea, Taiwan, Canada, Singapore, Australia / New Zealand
Patterns	Type 1 to 4 for Japan (W53 and W56); Type 1 to 6 for all other regions Short-pulse, long-pulse chirp, staggered-PRF and frequency-hopping patterns
Channel	Any 5 GHz Wi-Fi channel, 36 to 165 (5180 to 5825 MHz). Pattern and channel are independent.
Intended use	DFS development and pre-compliance testing
AWGN	
Centre frequency	2400 to 6000 MHz in 1 MHz steps
Noise bandwidth	2 to 20 MHz in 1 MHz steps
All non-Wi-Fi signals	
Output level setting	−35 to +12 dBm in 1 dB steps
Playback	Continuous loop until stopped
RF output	RF3 (front panel)

Notes

- The non-Wi-Fi output level is a control setting, not a calibrated value. The actual output varies with frequency and is lower towards 5 GHz; where the absolute level matters, measure it with a power meter.
- Five long-pulse chirp patterns (Japan Type 3; Korea, Taiwan, Canada and FCC Type 5) are listed but cannot be transmitted with the current firmware.
- The radar patterns are representative test signals for exercising radar detection. Formal certification requires accredited test equipment.

Control and connectivity

MT-Wave is operated from a built-in web control panel. Open the unit's address in a browser and both radios appear side by side, each with its own settings, live status and Apply / Start / Stop buttons. The same functions are available to test scripts over an HTTP API.



Web control panel: Wi-Fi interference (left) and non-Wi-Fi interference with Bluetooth, Zigbee, Radar and AWGN tabs (right)

Parameter	Specification
Control panel	Built-in web interface for Chrome, Edge, Firefox or Safari. No software to install.
Network	Ethernet. Control panel at http://<unit address>:7000
Automation	HTTP API with JSON requests and replies at http://<unit address>:7000/api, with separate endpoints for the Wi-Fi and non-Wi-Fi radios
Status read-back	Idle / Transmitting state with timer, and the active configuration of each radio
Unit identity	Serial number, firmware version, radio MAC and IP address shown in the top bar
Start-up	Ready about 30 seconds after power-on; no manual start needed

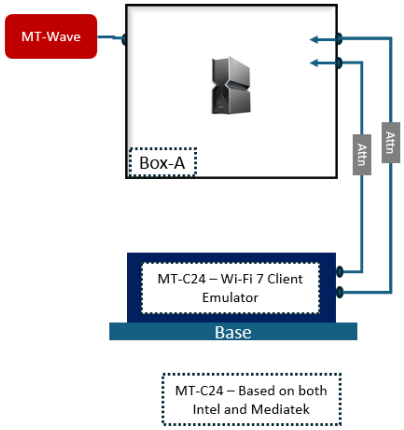
Front panel RF outputs



Each radio has its own SMA output on the front panel, so the two interference signals can be routed, attenuated or combined separately.

Output	Signal
RF1	Wi-Fi interference
RF3	Non-Wi-Fi interference: Bluetooth, Zigbee, Radar, AWGN

Reference test configuration



A typical Wavemetrik setup (MT-SYS1) pairs MT-Wave with the MT-C24 client emulator and a shielded enclosure. MT-Wave injects controlled interference while MT-C24 generates the reference traffic, giving repeatable rate-versus-range (RVR, RVRO) and coexistence runs.

- **MT-Wave**: interference source
- **Shielded enclosure (Box-A)**: holds the device under test
- **MT84-A**: programmable attenuators on the path under test
- **MT-C24**: client emulator and sniffer

Related products

Product	Description
MT-C24	Wi-Fi 7 client emulator with 24 or 48 client radios. MLO (EMLSR / MLMR), preamble puncturing and Multi-RU; 2×2 MIMO, up to 5 Gbps at 6 GHz; sniffing, band-steering and RVR / RVRO test automation.
MT84-A	Programmable attenuator: 95 dB range, 0.25 dB steps, ±0.5 dB accuracy.
MT-SYS1	Reference test system combining MT-Wave, MT-C24, MT84-A attenuators and a shielded enclosure.

Sample test results

The plots show device throughput for each WMM traffic class as the MT-Wave interferer level is stepped from −20 dBm (heavy interference) to −70 dBm (negligible interference), for two access points.

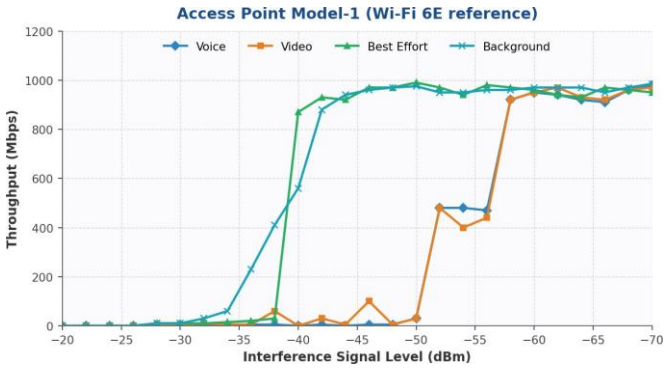


Figure 1. Access point Model-1 (Wi-Fi 6E). Best-effort and background traffic recover at about −40 dBm; voice and video recover at about −52 dBm.

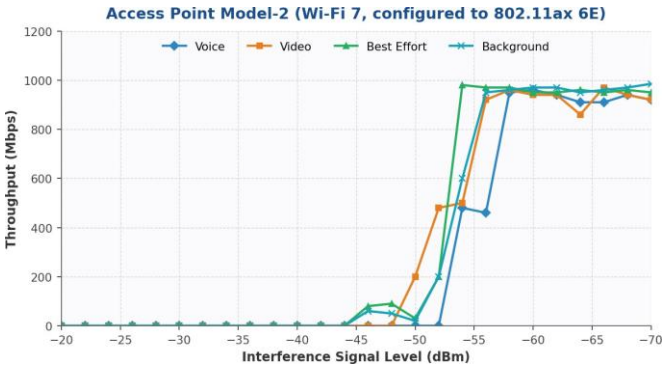


Figure 2. Access point Model-2 (Wi-Fi 7, set to 802.11ax 6E). All traffic classes need a weaker interferer, about −55 dBm, before throughput recovers.

Model-1 supports up to Wi-Fi 6E and Model-2 up to Wi-Fi 7; for a like-for-like comparison MT-C24 was set to 802.11ax on 6 GHz. Interferer levels include over-the-air and cable losses.

Wavemetrik Technologies Pvt. Ltd.

Bangalore, India | www.wavemetrik.com | sales@wavemetrik.com

Specifications subject to change without notice.